Examiner reports

Q1.

This integration question proved to be a good start for a large number of students with around 57% correctly choosing option 3, while around 32% chose option 2.

Q2.

This question was generally well attempted with over a quarter of students achieving full marks. Students recognised the need to expand $f'(x)$, integrate their expansion and evaluate the constant of integration from the given condition. Unfortunately the expansion proved to be too demanding algebraically for some, with incorrect squaring of the $2x$ and $\left(-\frac{3}{x}\right)$ terms the major issue.

The integration was generally well done, although some students had issues with the negative power and some omitted the constant of integration which prevented them from completing the solution.

Q3.

Almost all students selected the correct response for this multiple choice question. The most common incorrect answer was $\frac{dy}{dx} = -\frac{2}{x}$, but this was not significant in number.

Q9.

A majority of students could cope with the powers in part (a) but a significant proportion of others ended up with the wrong second term by thinking that the denominator x should be placed under both the x^2 and the $x^{1/2}$ terms resulting in a common error of the second term being given as $x^{1/2}$ a rather than $x^{3/2}$. Most students showed that they could differentiate accurately and a large majority showed that they knew the method for finding the equation of the normal to a curve, with many scoring at least 3 of the 4 marks in part (b)(ii). The final part of the question, which introduced the stationary point, usually gave average students the opportunity to pick up the first mark but in general only the most able students could then cope with the fractional indices to correctly reach $x^{5/2} = 8$ and then to solve this to show that the x-coordinate of the stationary point could be written as $2^{6/5}$.

Q10.

This question on fractional indices caused candidates more problems than expected.

In part (a), the two most common wrong answers for $\sqrt[4]{x^3}$ were $x^{\frac{4}{3}}$ and x^{12}. In part (b), a significant number of candidates failed to either split the numerator or multiply both its terms by x raised to the appropriate negative fractional power.

Despite the given form of the answer, $x^p - x^q$, it was not a rarity to see final answers left in the form $1 - x^2 - x^{\frac{3}{4}}$.

Q11.

Most candidates stated, or clearly indicated, the correct values for p, q and r in part (a) and appreciated that these answers could then be used in part (b). There were many correct solutions seen, although a number of candidates resorted to using logarithms rather than just the rules of powers. The most common error in applying laws of indices was to express $2^{\frac{1}{2}} \times 2^x$ as $2^{\frac{x}{2}}$ rather than $2^{\frac{1}{2}+x}$ which led to the common wrong answer $x = -6$. For full marks in part (b), candidates were required to state the value of x explicitly rather than leave the last line of their working as $2^x = 2^{-3.5}$.

Q12.

The majority of candidates answered part (a) successfully using the binomial expansion, Pascal's triangle or expanding brackets. Those candidates who wrote the expansion using the binomial coefficient notation from the formulae book did not always convince the examiner that they knew how to correctly evaluate the combinatorial coefficients.

In part (b)(i) very few candidates scored full marks with a large number of attempts seen where the numerator and denominator were integrated separately. Those candidates who realized the need for negative indices generally scored well. Most candidates picked up the method mark in the final part of the question for dealing with the limits in a correct manner. A few candidates just stated the correct answer for the definite integral without showing any working. Such an approach gained no credit as it did not explicitly indicate the use of 'Hence'.

Q13.

Most candidates answered parts (a) and (b) correctly; the most common incorrect values for k in part (a) were either $5\frac{1}{2}$ or $\frac{1}{5}$, and in part (b) it was disappointing to see 4 being integrated as $\frac{4^2}{2}$. The final part of the question was generally answered very badly with the incorrect answer, $y = 3x$, the equation of a line, seemingly appearing as often, if not more so, than the correct equation of the curve.

Q14.

Most candidates coped with the negative fractional powers well, differentiated correctly and found an equation of a normal, although there were a few equations of tangents presented as final answers to part (b)(ii). A common arithmetical slip in part (b)(ii) was '$\frac{1+\sqrt{1}}{1}$' incorrectly evaluated as 1. The expression for the second derivative was often correct in part (c)(i).

Part (c)(ii) was discriminating. The most popular approaches were to either put $x = 1$ in $\frac{d^2y}{dx^2}$, or to equate $\frac{d^2y}{dx^2}$ to zero and solve, for which no marks were scored.

A significant number of candidates did, however, earn some credit by making a more general comment on the sign of the second derivative but many lacked rigour and, in particular, assumed the second derivative has to be negative for a maximum, overlooking the possibility that the second derivative may be zero at a maximum.

Q15.

Most candidates were able to score marks in part (a), although the mark in part (a)(iii) was not awarded for answers left as '$\log_a 5^3$'. In part (b), candidates generally scored the method marks but frequently failed to pick up the accuracy mark. The most common wrong answer was '1.66', not always simply from a premature approximation.

Part (c), as expected, proved to be beyond the ability of the average candidate; some did score a mark for '$p = 2^m$', but this was often followed by '$pq = 2^m \times 8^n = 16^{mn}$'. Better candidates reached '$pq = 2^m \times 2^{3n}$', but a significant number of these candidates then simplified incorrectly to reach the wrong answer '$pq = 2^{3mn}$'.

Q16.

Most candidates gave the correct value for n in part (a). The most common wrong answer was '$n = -\frac{1}{4}$'.

The expansion of $\left(1+\frac{3}{x^2}\right)^2$ in part (b) caused more difficulty than expected. A common wrong answer was '$1 + 6x^{-2} + 9x^4$' although there were many other examples of incorrect answers offered.

In integrating $\left(1+\frac{3}{x^2}\right)^2$ in part (c) most candidates realised that their expansion in part (b) should be used but surprisingly the integration of 1 with respect to x caused just as many problems as the integration of the terms in x raised to negative powers. Although many candidates showed that they knew how to deal with the given limits, a lack of brackets in their numerical expressions frequently led to the wrong answer. The final mark was frequently lost because no **exact** answer (fraction or mixed number) was given.

Q17.

The work required to answer parts (a)(i) and (a)(ii) was well understood but a significant number of candidates did not give an explicit value for p or an explicit value for x. Candidates at this level should not be leaving their answers in an embedded form, for example, in part (a)(ii) answers were sometimes left as '52 × 0.75 =   125'.

Part (b) was answered better than the corresponding work in previous years and there continued to be an improvement in knowledge and use of the laws of logarithms as seen in solutions to part (c) where many candidates were able to score at least half marks by correctly applying two laws of logarithms. Only more able candidates either used $\log_a a = 1$ or applied $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ towards expressing x in terms of a correctly.

Q18.

Although many candidates were able to correctly write $\sqrt{x^3}$ in the form x^k it was not uncommon to see the incorrect answers k = 3.5 or k = $\frac{1}{3}$. Although follow through marks were available, a significant minority of candidates subsequently differentiated $x^{\frac{1}{3}}$ incorrectly as $\frac{1}{3}x^{\frac{2}{3}}$. In addition to arithmetical errors the most common error in finding the equation of the tangent was to use the gradient of the tangent as $-\frac{1}{5}$; in effect, the equation of the normal was found. Very few candidates failed to write their equation in the required form.

Q19.

A greater proportion of candidates than in previous years used the correct formulae when answering this topic, which involved an infinite geometric series. Many candidates found the correct value for the common ratio, although the incorrect value 1.25 was seen. Candidates who gave this incorrect value did not seem to be concerned that their answer for the sum to infinity was negative. In part (c) some candidates did not gain the accuracy mark because they only gave a one decimal place answer for the sum of the first twenty terms. The final part of the question, as expected, was a challenge to all. No credit was given to those who just used particular values for n. Better candidates stated that the nth term was 20 × 0.8^{n-1} but only the most able candidates gave the intermediate stage, 20 × 0.8^{-1} × 0.8n, or equivalent, to persuade the examiners that the printed result had been obtained convincingly.

Q20.

Although in part (a) most candidates gave the correct values for p, q and r, it was disappointing to see a significant number of candidates having to resort to a page of working involving logarithms to various bases before obtaining the three values. In part (b), many candidates preferred to use logarithms rather than use the laws of indices directly; both approaches were equally successful.

Q21.

Most candidates were able to correctly simplify the given expressions although $x^{\frac{9}{4}}$ was a common wrong answer in part (a)(iii) for the simplification of $\left(x^{\frac{3}{2}}\right)^2$. The majority of candidates displayed a good understanding of integration but a significant proportion of the entry left their answer to part (b)(i) in the form $\frac{3}{\frac{3}{2}}x^{\frac{3}{2}}$. The final mark of three required the answer to be simplified further and the constant of integration present. The majority of candidates showed sufficient working in part (b)(ii) to show that they had used their answer to part (b)(i) and so gained the method mark, but a significant minority failed to evaluate $2\times\ 9^{\frac{3}{2}}$ correctly.

Q22.

Parts (a) and (b) were answered very well and although the powers in answers to part (c) were nearly always correct, the coefficients were sometimes the reciprocal of the powers in part (b). Part (d) caused significantly more problems to the candidates. Although the method of solving a definite integral was well understood, many candidates resorted to using decimal approximations and, in general, candidates displayed a weakness in converting fractional powers of numbers back into surds.

Q23.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by $\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}$, and $\frac{\sqrt{x}^2}{2}$, were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word ‘translation’. Some candidates wrote ‘tr’ but this was not considered to be clear enough to distinguish between ‘transformation’ and ‘translation’. Common wrong answers involved ‘stretch’ and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q24.

Most candidates found one correct term but only a minority found both. The common incorrect answers are illustrated by ‘$x^8 - x^{-3}$ and ‘$x^5 - x^3$’Most candidates were able to differentiate their expression for f(x), but the method required to show that f is an increasing function was not well known. Many did not attempt it and many others tried to justify it by considering the relative size of f ′ (x) for two particular values of x. Others incorrectly stated that ‘f ′(2) > f (2) so f is increasing’ or used the second derivative. Those who considered the sign of f ′(x) for the given domain, $x > 0$, usually gained both marks in a line of working. Those candidates who realised that the gradient of the tangent was given by f ′(1) usually obtained a correct follow through answer for the gradient of the normal. It is worth noting that many candidates went on to find the equation of the normal. Although, of course, this further work was not penalised, it may have used up valuable time for some of the candidates.